Lab2: Binary Semantic Segmentation

TA 劉子齊 Jonathan

In this lab, we will explore the fascinating field of **semantic segmentation** using deep learning techniques. Specifically, we will focus on **binary semantic segmentation**, where each image pixel is classified as foreground (object of interest) or background.

Dataset

The **Oxford-IIIT Pet Dataset** consists of images of cats and dogs, annotated with pixel-level labels for the pet regions. We will work with this dataset to train and evaluate its segmentation models.

Tasks

1. **Data Preparation**:
   - Load and preprocess the Oxford-IIIT Pet Dataset.
   - Use the provided training and validation splits.
   - Preprocess the datasets to obtain better results.
2. **Model Architecture**:
   - Design 2 encoder-decoder networks.
   - The following two architectures are required:
     1. UNet
     2. ResNet34 + UNet (ResNet34 as the encoder and UNet as the decoder)
3. **Training and Evaluation**:
   - Train the model using appropriate loss functions (e.g., cross-entropy).
   - Monitor training progress and validate on the validation set.
   - Evaluate the model's performance using **Dice Score**.
4. **Inference**:
   - Apply the trained model to unseen images from the test set.
   - Submit predictions to Kaggle and obtain the Dice score from the leaderboard.

Important Date:

Code, Kaggle submission, and summary submission deadline: 3/31 (Tue.) 23:59 (UTC+8)

Late Submission Policy

Submissions made after the deadline but before **5/21 (Thu.) 23:59 (UTC+8)** will still be accepted but will not be included in the leaderboard ranking.

For late submissions:

- The Kaggle Results score will be capped at the baseline level

- That is, even if your ranking is high, the maximum score you can receive is equivalent to:
  - 70 pts per model (Dice > 0.85)

Submissions after 5/21 (Thu.) 23:59 (UTC+8) will not be accepted and will receive a grade of 0.

Submission Instructions

You must submit the following **three components**:

1. **Short Technical Summary (.txt)**
   - Filename: DL_Lab2_YourStudentID_name.txt
   - Example: DL_Lab2_311605004_劉子齊.txt
   - Maximum 2000 characters (including spaces)

2. **Source Code (.zip)**
   - Compress all your source code into a single zip file.
   - Filename: DL_Lab2_YourStudentID_name.zip
   - Example: DL_Lab2_311605004_劉子齊.zip

3. **Kaggle Submission**
   - Please set your team name as your StudentID_Name
     - Example: 311605004_劉子齊
     - I'll delete your team if you fail to do so.
   - Maximum Daily Submission Limit: 3
   - UNet: https://www.kaggle.com/t/e099deedbed94ea18629ff041a70ff87
   - ResNet34_UNet: https://www.kaggle.com/t/fa97cf1a0b6d49d89c73072c044c7b1f

Important Notes:

- Students must also submit predictions to the Kaggle leaderboard for both required models.
- You must complete all required components, including Kaggle submission, summary, and source code; otherwise, your lab will be graded as **0 points**.
- If the filename does not follow the specified naming rule, there will be a **10-point penalty**.
- If the TA is unable to unzip your code or open your summary file, it will be considered a missing submission and graded as 0 points.

Requirements:

1. Design your own data preprocessing method.

2. Implement the UNet & ResNet34_UNet architecture independently; **do not call the model from any library**.

3. Train your model from scratch; **do not load parameters** from any pre-trained model.

4. Implement your custom training, evaluating, and inferencing code.

Dataset - Oxford-IIIT Pet Dataset:

The **Oxford-IIIT Pet Dataset** is a widely used benchmark in computer vision research, particularly for semantic segmentation and object recognition tasks. The dataset was created for fine-grained image analysis, specifically focusing on pet images (cats and dogs). It is a valuable resource for developing and evaluating models that can understand and segment objects within images.

The dataset contains 7,349 images of pets (primarily cats and dogs). Each image is annotated with pixel-level labels, indicating the boundaries of the pet regions. Annotations include binary masks, where pixels belonging to the pet are labeled as foreground, and the rest are labeled as background.

The bounding box annotations provided in the dataset are for localization purposes only and should **not** be included as part of the foreground region in your segmentation task. Please strictly use the pixel-level segmentation mask (trimap) to define the foreground (pet region), and treat all other pixels as background.

Image

GT mask

Download Link: https://www.robots.ox.ac.uk/~vgg/data/pets/

Dataset Usage Guidelines

- It is essential to keep the training, validation, and testing datasets completely separate.
- The **training and validation** sets should only be used for hyperparameter tuning and model evaluation during the training process.
- The **testing** set is reserved solely for final inference and must not be utilized at any point during training or validation. The ground truth labels for the test set are not provided.

- Please only train & validate your model on the training & validating dataset. TAs will retrain your models and review your code. If any use of the testing set is found during the training or validation phases, you will receive **0 points** for this lab.

- Students must train and validate models only on the provided training/validation split.

- Using test labels, reconstructing test masks from external sources, or any form of data leakage is strictly prohibited.

- TAs will randomly retrain and verify submissions.

- If submitted results cannot be reproduced, the lab will receive 0 points.

Binary Mask Definition (Important)

The provided annotations are trimaps with three labels:

- 1: Foreground (pet)
- 2: Background
- 3: Boundary

For this lab, the trimap must be converted into a binary mask:

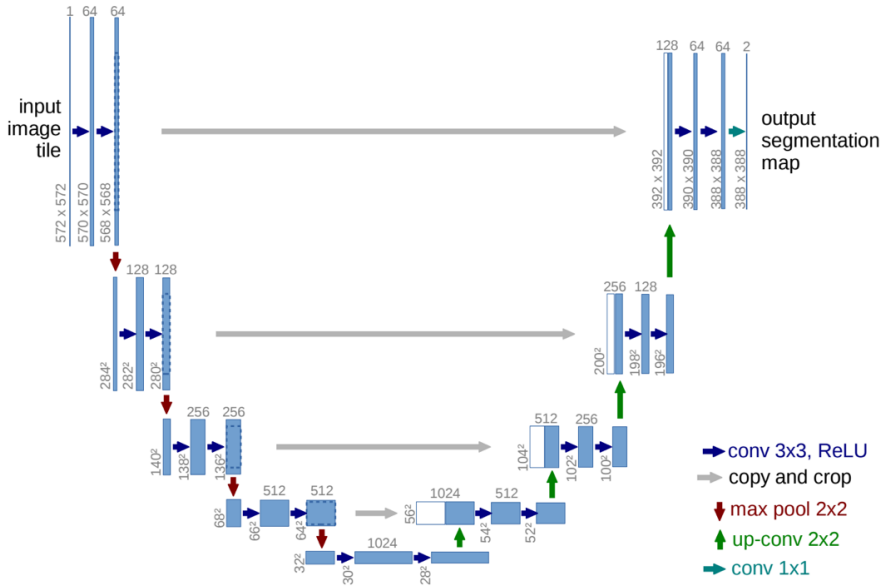
- Pixels with value 1 → Foreground (1)
- Pixels with value 2 or 3 → Background (0)

Boundary pixels must be treated as background.

Submissions that use different label mappings (e.g., treating boundary as foreground or ignoring boundary pixels) will be considered incorrect.

Model Architecture:

UNet



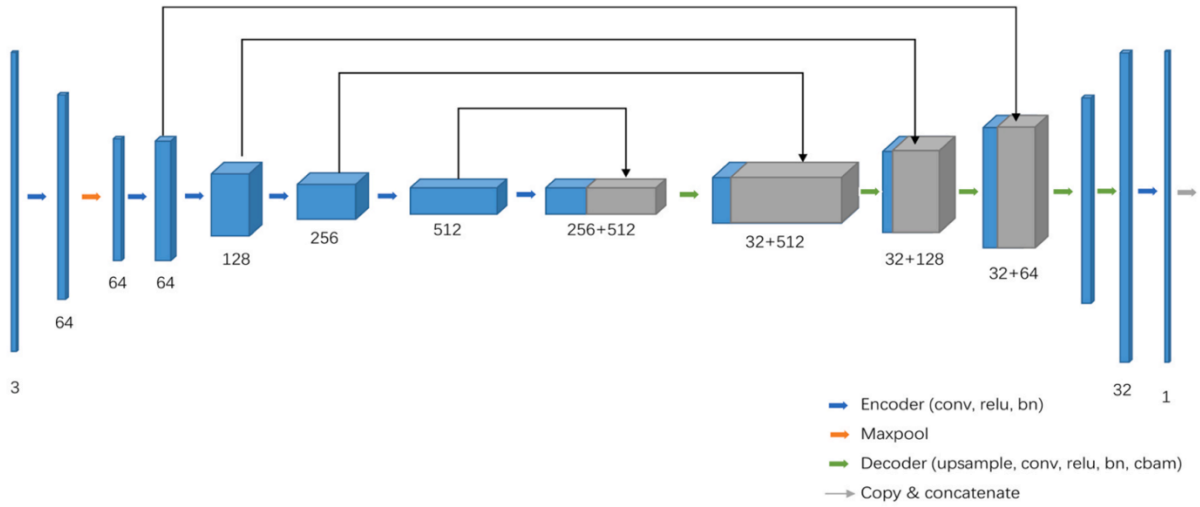
Reference: https://arxiv.org/abs/1505.04597v1

ResNet34

layer name	output size	18-layer	34-layer	50-layer	101-layer	152-layer
conv1	112×112	7×7, 64, stride 2				
conv2_x	56×56	3×3 max pool, stride 2				
		$\begin{bmatrix} 3\times3, 64 \\ 3\times3, 64 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 64 \\ 3\times3, 64 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix}\times3$
conv3_x	28×28	$\begin{bmatrix} 3\times3, 128 \\ 3\times3, 128 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 128 \\ 3\times3, 128 \end{bmatrix}\times4$	$\begin{bmatrix} 1\times1, 128 \\ 3\times3, 128 \\ 1\times1, 512 \end{bmatrix}\times4$	$\begin{bmatrix} 1\times1, 128 \\ 3\times3, 128 \\ 1\times1, 512 \end{bmatrix}\times4$	$\begin{bmatrix} 1\times1, 128 \\ 3\times3, 128 \\ 1\times1, 512 \end{bmatrix}\times8$
conv4_x	14×14	$\begin{bmatrix} 3\times3, 256 \\ 3\times3, 256 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 256 \\ 3\times3, 256 \end{bmatrix}\times6$	$\begin{bmatrix} 1\times1, 256 \\ 3\times3, 256 \\ 1\times1, 1024 \end{bmatrix}\times6$	$\begin{bmatrix} 1\times1, 256 \\ 3\times3, 256 \\ 1\times1, 1024 \end{bmatrix}\times23$	$\begin{bmatrix} 1\times1, 256 \\ 3\times3, 256 \\ 1\times1, 1024 \end{bmatrix}\times36$
conv5_x	7×7	$\begin{bmatrix} 3\times3, 512 \\ 3\times3, 512 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 512 \\ 3\times3, 512 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 512 \\ 3\times3, 512 \\ 1\times1, 2048 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 512 \\ 3\times3, 512 \\ 1\times1, 2048 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 512 \\ 3\times3, 512 \\ 1\times1, 2048 \end{bmatrix}\times3$
	1×1	average pool, 1000-d fc, softmax				
FLOPs		1.8×10^9	3.6×10^9	3.8×10^9	7.6×10^9	11.3×10^9

Reference: https://arxiv.org/abs/1512.03385

ResNet34 (Encoder) + UNet (Decoder)



Reference: https://www.researchgate.net/publication/359463249_Deep_learning-based_pelvic_levator_hiatus_segmentation_from_ultrasound_images

Implementation Details:

File Structure

Please organize your project files using the following directory structure that separates data, code, and other resources. It is strongly recommended to follow the following file structure.

```
├── dataset/
│   └── oxford-iiit-pet/
├── src/
│   ├── models/
│   │   ├── unet.py
│   │   └── resnet34_unet.py
│   ├── oxford_pet.py
│   ├── utils.py
│   ├── train.py
│   ├── evaluate.py
│   └── inference.py
├── saved_models/
└── requirements.txt
```

dataset/
This directory contains data related to the **Oxford-IIIT Pet Dataset**. Inside dataset/, there is a

subdirectory named oxford-iiit-pet/. The dataset files (images, annotations, etc.) are likely stored within this subdirectory. We access and manipulate the dataset from here.

src/
The src/ directory holds the source code for your project. It contains various Python files responsible for different functionalities. Let's explore the contents of src/:

- **models/**: This subdirectory likely contains implementations of different neural network architectures. Specifically, there are two files:
  - unet.py: Contains the UNet architecture implementation.
  - resnet34_unet.py: Combines the ResNet-34 backbone with the UNet architecture.

- **oxford_pet.py**: This file contains code for handling the Oxford-IIIT Pet Dataset. It includes data loading, preprocessing, and other dataset-related functions.

- **utils.py**: The utils.py file typically contains utility functions used across the project. These functions might include visualization tools, metrics computation, or other common tasks.

- **train.py**: This file contains code for training the neural network models. It includes functions related to model training, optimization, and backpropagation.

- **evaluate.py**: This file probably handles model evaluation. It includes functions to assess model performance on validation.

- **inference.py**: This file deals with model inference (making predictions) on unseen data. It includes functions to apply the trained model to new images.

saved_models/
The saved_models/ directory is where you would store trained model checkpoints. After training, the best-performing model weights are saved here for later use. Please save the trained model weights in the ".pth" files.

requirements.txt (Required)
You must include a requirements.txt file in your submission. This file should list all dependencies required to rebuild the training environment for retraining your code.
Failure to provide a valid and functional requirements.txt file will result in a score of 0 for this lab.

Inference Code Organization Guidelines

- All testing (inference) logic must be placed in the inference.py file.

- Do not include any test-time inference code in train.py or evaluate.py.

- The TAs will run your code starting from inference.py for testing purposes; if your inference logic is missing or scattered in other files, it will be considered incomplete, and you will receive 0 points for the testing phase.

- This requirement is in place to ensure that your code is well organized and reproducible.

Dice Score

The Dice score, also called the Dice Similarity Coefficient, serves as a metric for assessing the similarity between two sets of data, often represented as binary arrays. The Dice score quantifies how closely a predicted segmentation mask aligns with the ground truth segmentation mask in image segmentation. Its range spans from 0 (indicating no overlap) to 1 (representing perfect alignment).

Mathematically, the Dice score is computed as:

Dice score = 2 * (number of common pixels) / (predicted img size + ground truth img size)

In simpler terms, the Dice score corresponds to twice the size of the intersection divided by the sum of the sizes of the two sets. It measures the proportion of overlap normalized by the overall set sizes.

Reference: https://oecd.ai/en/catalogue/metrics/dice-score

Dice Score Calculation Guidelines

- The Dice score for each model must be evaluated using the entire **testing dataset**.

- You are required to run inference on every image in the test set and then compute the average Dice score across all images.

- Do not report the Dice score from only the best-performing sample as the model's overall Dice score.

- The **average Dice score for the entire testing set** will serve as the final metric for grading your experimental results.

Scoring

Final score = Kaggle Results (90%) + Short Technical Summary (10%)

Please note that I will randomly select students to perform a demo. This demo may result in an additional score adjustment of up to 30 points, either added or deducted.

We will notify selected students via email, so please check your inbox regularly. If a selected student does not respond within 3 days or fails to participate in the demo as scheduled, Lab 2 will be graded as zero.

Kaggle Results (90%)

- UNet leaderboard: 45%
- ResNet34_UNet leaderboard: 45%

For each model:

- Top 10%: 100 pts
- Top 25%: 90 pts
- Top 50%: 85 pts
- Top 75%: 80 pts
- Others above baseline (Dice > 0.85): 70 pts
- Dice ≤ 0.85: 0 pts

Ranking is determined based on the final leaderboard at the deadline. Only the best submission before the deadline will be considered.

The test set must not be used for hyperparameter tuning or model selection. Any attempt to overfit specifically to the test set through excessive submissions or manual tuning based on leaderboard feedback is discouraged and may be penalized.

Short Technical Summary (10%)

- File format: .txt
- Maximum length: 2000 characters (including spaces)
- Content should include:
  - implementation highlights
  - data preprocessing strategy
  - training strategy
  - key observations
  - comparison between UNet and ResNet34_UNet

Reproducibility Requirement

Your code must be executable with a single command and should not require manual intervention.

Other Rules

If the zip filename or the report has a format error, it will be a penalty (-10 points).

If you're randomly picked to attend the demo session, you need to load the trained model and show the results of the two models in the demo phase. If you fail to do so, you will get 0 points on the model that you couldn't successfully demonstrate.

Please ensure the TA can successfully retrain your model using the uploaded code. If the TA is unable to do so, you will receive 0 points for this lab.

If you do not upload a requirements.txt file, or if the TA cannot successfully rebuild the environment using the provided file, you will receive 0 points for this lab.

If you're found faking your model predicting results, you'll get 0 points in this lab. Also, we will take disciplinary action under school regulations.

Plagiarism is strictly prohibited. If you're found guilty of plagiarizing another's code, you and the person/people you plagiarized will get 0 points in this lab. Also, we will take disciplinary action under school regulations.

If any issues are identified in your submitted files for this lab, the TA will reach out to you via email. Please monitor your inbox closely. Suppose the TA has not received your response within three days of emailing. In that case, it will be considered that you have forfeited the opportunity to provide further clarification and have accepted the corresponding penalty or consequences.

Academic Honesty and Use of AI Tools

- If you choose to use any AI language models (LLMs), such as ChatGPT, Copilot, or other generative tools to assist with writing your summary or code, you must clearly disclose this in your submission. Failure to disclose the use of these tools will result in a score of 0 points for the lab.

- Additionally, even if you do disclose your use of AI assistance, if your submission is determined to be primarily generated by AI (i.e., showing little to no original contribution or understanding), you will still receive 0 points for the lab.

- All students are responsible for upholding academic honesty and must ensure that their work reflects their own understanding and effort.